Queries to AIs_20260518(一)

20260518_00

I've successfully installed Linux in the Samsung notebook. I'll use it to set up Git, GitHub, VScode ... etc. Please tell me the detailed steps to accomplish them?

Excellent — now that Linux is installed, let's set up your development environment with Git, GitHub, and VS Code. Here's a clear step-by-step guide you can follow:

01

Update Your System

Keep your Linux installation secure and up to date before adding tools.

Open Terminal

- Run: `sudo apt update && sudo apt upgrade`
- Reboot if kernel updates were applied

02

Install Git

Git is essential for version control and connecting to GitHub.

Terminal command

- Run: `sudo apt install git`
- Verify with: `git --version`

03

Configure Git Identity

Set your name and email for commits.

Terminal command

- Run: `git config --global user.name "Your Name"`
- Run: `git config --global user.email "your@email.com"`
- Check with: `git config --list`

04

Generate SSH Key

SSH keys allow secure authentication with GitHub.

Terminal command

- Run: `ssh-keygen -t ed25519 -C "your@email.com"`
- Press Enter to accept default file location
- Add key to agent: `eval "$(ssh-agent -s)"` then `ssh-add ~/.ssh/id_ed25519`

05

Add SSH Key to GitHub

Link your local machine to your GitHub account.

- Copy public key: `cat ~/.ssh/id_ed25519.pub`
- Log in to GitHub → Settings → SSH and GPG keys → New SSH key
- Paste the key and save

06

Install VS Code

VS Code is a popular editor for web development.

Terminal command

- Run: `sudo snap install code --classic`
- Launch with: `code`
- Install extensions: Python, Django, GitHub Copilot (optional)

07

Test GitHub Connection

Verify that your SSH setup works.

Terminal command

- Run: `ssh -T git@github.com`
- Expected message: "Hi username! You've successfully authenticated..."

08

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Clone or Create a Repository

Start working with GitHub projects.

- Clone: `git clone git@github.com:username/repo.git`
- Or create a new repo on GitHub, then `git init` locally
- Add remote: `git remote add origin git@github.com:username/repo.git`

✅ With this setup, you'll be ready to start coding, commit changes, and push them to GitHub directly from your Linux notebook.

Would you like me to also show you **how to set up a Django project** in VS Code so you can immediately begin web application development?

Queries to AIs_20260518(一)

20260518_01

What is the purpose of "Add key to agent" in step (4) above?

Great question — this is an important detail in the Git + SSH setup process.

🔑 Purpose of "Add key to agent"

When you generate an SSH key pair, you get:

- **Private key** → stored securely on your computer.
- **Public key** → uploaded to GitHub.

Normally, every time you use Git over SSH, the system needs access to your private key. If the key is **passphrase-protected**, you'd have to type the passphrase each time. That's inconvenient.

The **SSH agent** is a background program that:

- Stores your private key in memory after you unlock it once.
- Provides the key automatically to SSH/Git when needed.
- Saves you from re-entering the passphrase every time you push/pull.

So when you run:

bash
```
ssh-add ~/.ssh/id_ed25519
```

you're telling the agent: *"Keep this key ready for me."* From then on, GitHub authentication works seamlessly without repeated prompts.

⚙️ Why it matters

- **Convenience**: No need to type your passphrase constantly.
- **Security**: Your private key stays in memory, not exposed in files or typed repeatedly.
- **Workflow**: Makes Git operations (clone, push, pull) smooth and automatic.

👉 In short: *"Add key to agent" is about loading your SSH private key into a helper program so Git can use it automatically without bothering you each time.*

Would you like me to show you how to **set up the agent to start automatically on login** so you don't have to run `ssh-add` manually every time you reboot?